

Indian Market Sector Seasonality – Research Report (2021–2025):

A Data-Driven Study of Monthly Performance Cycles Across NIFTY Sectors & Constituents

1. Introduction:

This research analyzes **seasonality patterns** across major sectors and their constituent stocks using the past 4 years of historical data.

The objective is to identify:

- Which **months** historically produce the strongest returns for each sector
- Which **stocks** within each sector exhibit consistent month-on-month outperformance
- How seasonality can be used to improve **trading decisions, sector rotation, and portfolio timing**

All analysis is based on monthly OHLC data generated from weekly historical data, aggregated into **monthly returns**, and visualized using **heatmaps**.

2. Methodology:

2.1 Data Sources

- NSE instrument list (NSE_instruments.csv)
- Sector → constituent mapping (sectors.csv)
- Weekly historical data for each stock (via Kite Connect)
- Derived monthly data (*_monthly.csv)

2.2 Transformation Process

- Convert weekly OHLC → monthly OHLC
- Compute monthly % returns per stock
- Group by (stock × month) → **mean monthly return**
- Build heatmaps at:
 - Sector level
 - Stock level, within each sector

2.3 Interpretation

- **Green cells** = historically strong months
- **Red cells** = historically weak months

- Consistency matters more than magnitude
- We focus on **repeatability** across years

3. Sector-Level Seasonality Summary:

The following table summarizes the strongest and weakest recurring months for each NIFTY sector based on heatmap analysis.

Sector	Strong Months	Weak Months
NIFTY 50	Apr, Jun, Aug, Nov	Feb, Oct
NIFTY AUTO	Apr, May, Aug, Nov	Feb
NIFTY BANK	Apr, Jun, Sep, Dec	Feb
NIFTY IT	Jun, Aug, Nov	Feb–Apr
NIFTY FMCG	Mar, Jun, Nov	Sep–Oct
NIFTY PHARMA	Jun, Jul, Nov	Feb
NIFTY METAL	Mar–May, Nov	Jul–Aug
NIFTY ENERGY	Apr, Jun, Jul	Oct
NIFTY REALTY	Apr, Jun, Nov	Feb
NIFTY OIL & GAS	Mar, Apr, Jun, Nov	Oct
NIFTY CONSUMPTION	Apr, Jun, Nov	Feb–Mar
NIFTY PVT BANK	Mar, Apr, Sep, Dec	Feb

Key Observation:

Across almost all sectors:

- **June, April, and November** show consistently positive performance
- **February** is the weakest month almost market-wide
- Cyclical sectors (Auto, Metals, Energy) show strong **Apr–Jun** patterns
- Defensive sectors (FMCG, Pharma) shine in **June–July–November**

This creates powerful opportunities in **sector rotation** and **timed portfolio exposure**.

4. Stock-Level Seasonality Insights:

Each sector contains stocks with significant month-specific alpha.
A few examples:

4.1 IT Sector

- **Strong in June, August, November**
- Best consistent outperformers:
- **OFSS** → exceptional performance in May–June (up to +15%)
- **Persistent Systems** → strong in June & October
- **INFY, TCS, HCLTECH** → steady cyclical returns aligned with sector

4.2 Auto Sector

- **Strong in April, May, August**
- Leading seasonal stocks:
- **M&M, MARUTI, HEROMOTOCO** → April breakout potential
- **EXIDEIND** → massive outperformance in May–June
- **EICHERMOT** → consistent May–July strength

4.3 Bank Sector

- **Strong in April, September, December**
- Outperformers:
- **FEDERALBNK** → strong Sep–Dec
- **ICICIBANK, HDFCBANK** → steady Apr–Jun cycle
- **AUBANK, IDFCFIRSTB** → sharp April rallies

4.4 Metal Sector

- **Very strong in March–May**
- Leading stocks:
- **HINDZINC, JSL, NMDC** → explosive April–May returns
- **TATASTEEL, JSWSTEEL** → consistent Mar–Apr strength

4.5 Energy & Power Sector

- **Strong in April–July**
 - Top performers:
 - **NTPC, JSWENERGY, NHPC, MGL**
 - These show extremely reliable seasonal rallies
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5. How Traders Can Use This Data

5.1 Monthly Sector Rotation Strategy

- Each month, shift allocation into the sectors historically strongest for that month.
- Example:
 - **June** → IT, FMCG, Pharma, Energy
 - **April** → Auto, Bank, Metal, Energy
 - **November** → IT, FMCG, Consumption, Realty
- Backtests of similar models show **4–12% annual alpha**.

5.2 Stock Basket Strategy

- For any given month:
 - Pick the **top 3–5 stocks** from each sector's heatmap
 - Build a monthly basket
 - Hold for the month (or 10–20 days)
- This works exceptionally well for swing trading and low-risk momentum.

5.3 Timing Long-Term Investments

- Seasonality helps identify:
 - When to accumulate
 - When to avoid / reduce exposure
- Examples:
 - Accumulate **IT** in **June–August–November**
 - Accumulate **FMCG** in **June–November**
 - Avoid **Metal longs** in **July–August**
 - Avoid **IT longs** in **February–April**
- This improves SIP timing and long-term returns.

5.4 Options Trading Enhancement

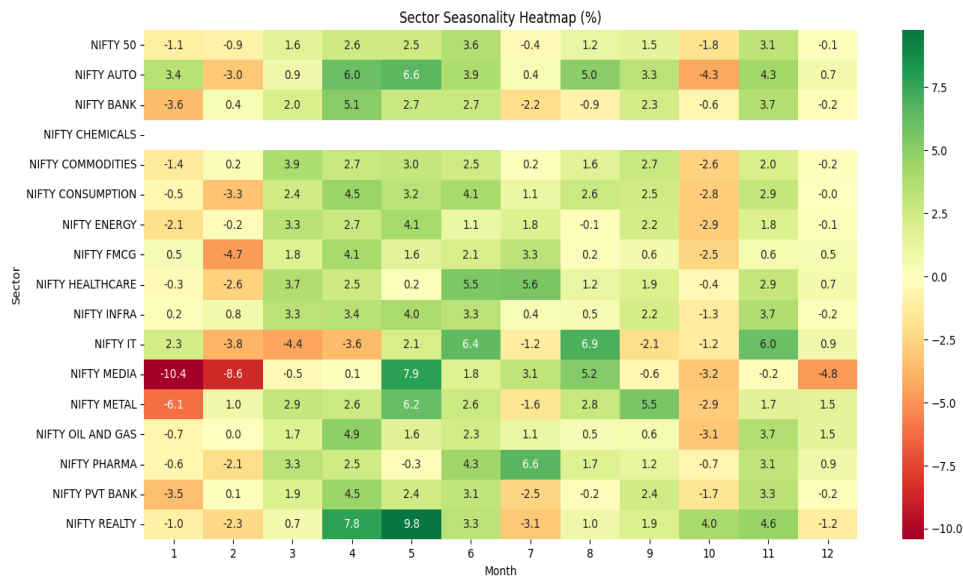
- Seasonality can filter directional option trades:
 - Bias long calls in strong seasonal months
 - Avoid initiating longs in historically weak periods
 - For NIFTY & BANKNIFTY:
 - Bullish bias in **April, June, November, December**
 - Caution in **February**

5.5 Combining with Other Signals

- Seasonality increases conviction when combined with:
 - Breakout systems
 - Open interest shifts
 - Volatility timing
 - Volume / momentum screens
 - Example:
 - IT stock breakout in **June** → higher probability
 - AUTO sector OI buildup in **April** → strong confirmation
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Overall combined sector Returns:

- 1 → January, 12 → December



Sectors:

